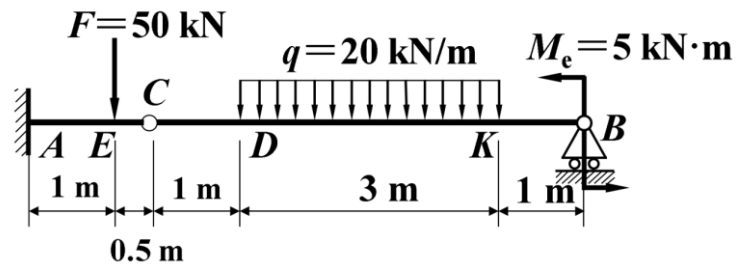


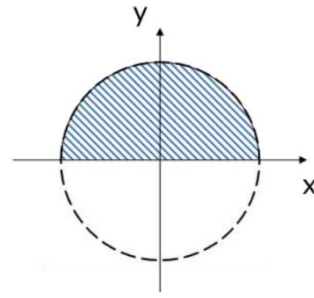
Submission: Please submit the scanned/typed document to Canvas.

1. Draw the shear-force diagram and bending-moment diagram for the following combination beam.



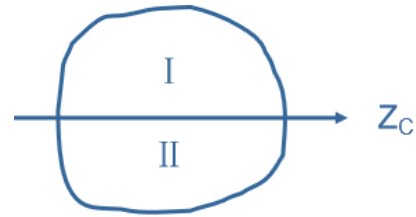
2. For the semicircle shown below, if the circle center is located at the coordinate origin, then ()

- A. $S_x = S_y, I_x \neq I_y$
 B. $S_x = S_y, I_x = I_y$
 C. $S_x \neq S_y, I_x \neq I_y$
 D. $S_x \neq S_y, I_x = I_y$

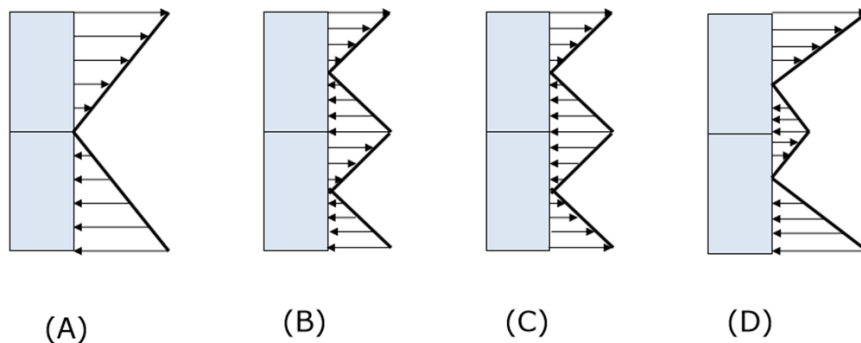
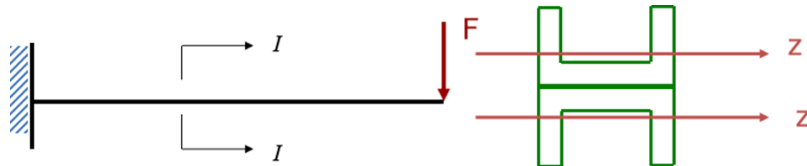


3. For the arbitrary shaped cross-section shown, one centroidal axis z_c (Passing through the centroid) divides the section into two parts I and II. Among the following formulas, () must always hold.

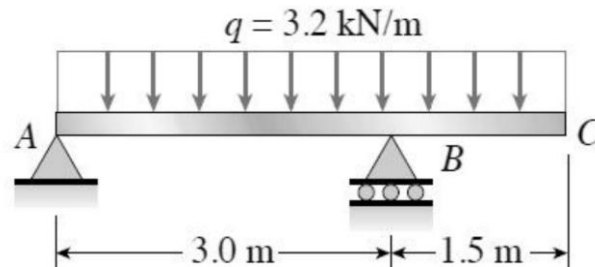
- A. $I_{z_c}^I + I_{z_c}^{II} = 0$
 B. $I_{z_c}^I - I_{z_c}^{II} = 0$
 C. $S_{z_c}^I + S_{z_c}^{II} = 0$
 D. $A^I = A^{II}$



4. A cantilever beam is formed by placing two identical C-channel beams back-to-back (without any connection between them). The overall cross section is shown in the figure below. Under the applied load F , the normal stress distribution on the cross-section is as illustrated in () (ignore the stress variation in the out-of-plane direction).



5. An overhanging C-channel beam ABC is applied with a uniform load whose intensity is $q = 3.2 \text{ kN/m}$. The cross-section of C-channel shape $b = 300 \text{ mm}$, $h = 80 \text{ mm}$, and the web thickness $t = 12 \text{ mm}$. Determine $\sigma_{\max}$ of both tensile and compressive stress due to the uniform load. (Hint: The difficulty of this problem lies in calculating the moment of inertia of the C-channel beam. Think about the parallel axis theorem.)



Cross section:

